$$u(l) = \frac{0.001}{\sqrt{12}} = 0.000289m$$
$$u(D) = \frac{0.00002}{\sqrt{12}} = 0.000006m$$
$$----$$
$$\bar{d} = \frac{1}{10}\sum_{i=1}^{10} d_i$$
$$\bar{d} = 1.18mm = 0.001182m$$

$$u_A(d) = \sqrt{\frac{1}{90}\sum_{i=1}^{10}(d_i - \bar{d})^2}$$
$$u_A(d) = 0.00000290593m$$

$$u_B(d) = \frac{0.00001}{\sqrt{12}} = 0.000003m$$

$$u(d) = \sqrt{u_A^2(d) + u_B^2(d)}$$
$$u(d) = \sqrt{0.00000290593^2 + 0.000003^2}$$
$$u(d) = 0.00000417665m$$
$$----$$
$$100T_k = 551.5s$$
$$551.2s < 100T_k < 551.8s$$
$$5.512s < T_k < 5.518s$$
$$T_k = 5.515s$$
$$u(T_k) = 0.003s$$

$$J = 0.5mR^2$$
$$J = 0.5 \cdot 6.22kg \cdot 0.13025^2 m$$
$$J = 0.052761kg \cdot m^2$$

$$u(J) = \sqrt{(\frac{R^2}{2})^2 u^2(m) + (mR)^2 u^2(R)}$$
$$u(J) = \sqrt{(\frac{0.13025^2 m}{2})^2 \cdot 0 + (6.22kg \cdot 0.13025)^2 \cdot 0.000003^2}$$
$$u(J) = 0.000002430465kg \cdot m^2$$
$$----$$
$$G = \frac{32\pi l J}{d^4 T_k^2}$$
$$G = \frac{32\pi 0.88m \cdot 0.052761kg \cdot m^2}{0.001182^4 m \cdot 5.515^2 s}$$
$$G = 7.86203 \cdot 10^{10} J \cdot s$$

$$u(G) = \sqrt{(\frac{\partial G}{\partial l})^2 u^2(l) + (\frac{\partial G}{\partial J})^2 u^2(J) + (\frac{\partial G}{\partial d})^2 u^2(d) + (\frac{\partial G}{\partial T_k})^2 u^2(T_k)}$$
$$u(G) = \sqrt{(\frac{32\pi J}{d^4 T_k^2})^2 u^2(l) + (\frac{32\pi l}{d^4 T_k^2})^2 u^2(J) + (\frac{-128\pi J l}{d^5 T_k^2})^2 u^2(d) + (\frac{-64\pi J l}{d^4 T_k^3})^2 u^2(T_k)}$$
$$u(G) = \sqrt{(\frac{32\pi \cdot 0.052761}{0.001182^4 \cdot 5.515^2})^2 \cdot 0.000289^2 + (\frac{32\pi \cdot 0.88}{0.001182^4 \cdot 5.515^2})^2 \cdot 0.000002430465^2 + (\frac{-128\pi \cdot 0.052761 \cdot 0.88}{0.001182^5 \cdot 5.515^2})^2 \cdot 0.00000417665^2 + (\frac{-64\pi \cdot 0.052761 \cdot 0.88}{0.001182^4 \cdot 5.515^3})^2 \cdot 0.003^2}$$
$$u(G) = \sqrt{6.66653 \cdot 10^{14} + 1.31166 \cdot 10^{13} + 1.23484 \cdot 10^{18} + 7.31612 \cdot 10^{15}}$$
$$u(G) = 1.11483 \cdot 10^9 J \cdot s$$

$$\mathbf{G = (7.86 \pm 0.11) \cdot 10^{10} J \cdot s}$$

$$100T_k = 93.2s$$
$$T_k = 0.932s$$
$$----$$
$$J_r = \frac{d^4T_k^2G}{32\pi l}$$
$$J_r = \frac{0.001182^4m \cdot 0.932^2s \cdot 7.86203 \cdot 10^{10}J \cdot s}{32\pi 0.88m}$$
$$J_r = 0.001507kg \cdot m^2 = 1.507g \cdot m^2$$

$$u(J_r) = \sqrt{(\frac{\partial J_k}{\partial d})^2u^2(d) + (\frac{\partial J_k}{\partial T_k})^2u^2(T_k) + (\frac{\partial J_k}{\partial G})^2u^2(G) + (\frac{\partial J_k}{\partial l})^2u^2(l)}$$
$$u(J_r) = \sqrt{(\frac{d^3T_k^2G}{8\pi l})^2u^2(d) + (\frac{d^4T_kG}{16\pi l})^2u^2(T_k) + (\frac{d^4T_k^2}{32\pi l})^2u^2(G) + (\frac{-d^4T_k^2G}{32\pi l^2})^2u^2(l)}$$
$$u(J_r) = \sqrt{\begin{array}{l}(\frac{0.001182^3 0.932^2 7.86203 \cdot 10^{10}}{8\pi 0.88})^2 \cdot 0.00000417665^2 + \\ (\frac{0.001182^4 0.9327.86203 \cdot 10^{10}}{16\pi 0.88})^2 \cdot 0.003^2 + (\frac{0.001182^4 0.932^2}{32\pi 0.88})^2 \cdot (1.11483 \cdot 10^9)^2 + \\ (\frac{-0.001182^4 0.932^2 7.86203 \cdot 10^{10}}{32\pi 0.88^2})^2 \cdot 0.000289^2\end{array}}$$
$$u(J_r) = \sqrt{4.53575 \cdot 10^{-10} + 9.40976 \cdot 10^{-11} + 4.56516 \cdot 10^{-10} + 2.44871 \cdot 10^{-13}}$$
$$u(J_r) = 0.0000316928kg \cdot m^2 = 0.0316928g \cdot m^2$$

$$\mathbf{J_r = (1.507 \pm 0.032)g \cdot m^2}$$